数学物理方法 · 第 14 讲
Mathematical Methods for Physics
Lecture 14

δ 函数、广义傅里叶变换及其性质

The Delta Function, Generalized Fourier Transforms, and Their Properties

7.3 δ 函数与广义傅里叶变换
7.3 The Delta Function and Generalized Fourier Transforms

1. δ 函数与矩形脉冲近似
The Delta Function and Its Rectangular-Pulse Approximation

δ 函数又叫单位脉冲函数。下式作直观描述：

The δ function is also called the unit impulse. The original notes give the following intuitive description.

$$\delta(x)=\begin{cases}0, & x\neq 0,\\ \infty, & x=0,\end{cases}\qquad \int_{-\infty}^{+\infty}\delta(x)\,\mathrm{d}x=1.$$

定义 1

Definition 1

$$\delta(x)=\lim_{\tau\to 0}\delta_\tau(x),$$

其中

where

$$\delta_\tau(x)=\begin{cases}\dfrac{1}{\tau}, & |x|\le\dfrac{\tau}{2},\\ 0, & |x|>\dfrac{\tau}{2}.\end{cases}$$

原图说明：矩形脉冲以原点为中心，宽度为 τ，高度为 $1/\tau$，面积为 1；当 $\tau\to 0$ 时，脉冲越来越窄、越来越高。$\delta(x)$ 用原点处的竖直箭头表示，旁标 “1” 表示单位积分。

Description of the original figure. The rectangular pulse is centered at the origin, has width τ and height $1/\tau$, and has unit area. As $\tau\to 0$, it becomes narrower and taller. The delta function is represented by a vertical arrow at the origin; the label “1” denotes its unit integral.

Notes:

2. δ 函数的泛函定义

The Delta Function as a Functional

定义 2（数学定义）

Definition 2 (mathematical definition)

$\delta(t)$ 是一个“泛函”，满足

The delta function $\delta(t)$ is a functional satisfying

$$\int_{-\infty}^{+\infty} \delta(t-t_0)\varphi(t)\,\mathrm{d}t = \varphi(t_0), \qquad \forall \varphi \in C(\mathbb{R}).$$

其导数满足

Its derivative satisfies

$$\int_{-\infty}^{+\infty} \delta'(t-t_0)\varphi(t)\,\mathrm{d}t = -\varphi'(t_0), \qquad \forall \varphi \in C^1(\mathbb{R}).$$

其 n 阶导数满足

Its nth derivative satisfies

$$\int_{-\infty}^{+\infty} \delta^{(n)}(t-t_0)\varphi(t)\,\mathrm{d}t = (-1)^n\varphi^{(n)}(t_0), \qquad \forall \varphi \in C^n(\mathbb{R}).$$

Notes:

3. δ 函数的傅里叶变换与积分表示

Fourier Transform and Integral Representation of the Delta Function

例 1：$\delta(t)$ 的傅里叶变换称为广义傅里叶变换。

Example 1. The Fourier transform of $\delta(t)$ is understood as a generalized Fourier transform.

$$\mathcal{F}[\delta(t)] = \int_{-\infty}^{+\infty} \delta(t)e^{-i\omega t}\,\mathrm{d}t = e^{-i\omega t}\big|_{t=0} = 1.$$

因此 $\{\delta(t), 1\}$ 构成一个傅里叶变换对。由反变换，

Thus $\{\delta(t), 1\}$ is a Fourier transform pair. By the inverse transform,

$$\delta(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} 1 \cdot e^{i\omega t}\,\mathrm{d}\omega.$$

定义 3（$\delta(t)$ 的积分定义）

Definition 3 (integral representation of $\delta(t)$)

$$\delta(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} e^{i\omega t}\,\mathrm{d}\omega.$$

注
Note

$$\int_{-\infty}^{+\infty} e^{\pm i\omega t}\,\mathrm{d}t = 2\pi\delta(\omega). \tag{*}$$

整理注：上述积分等式在广义函数意义下理解。
Editorial note. These integral identities are interpreted in the sense of distributions.

教材 P.340：常用广义傅里叶变换对。原稿首先列出第 10 对 $\{\delta(t), 1\}$，并提示第 12、13、14、15 对。
Textbook, p. 340: common generalized Fourier transform pairs. The original notes first list pair 10, $\{\delta(t), 1\}$, and then refer to pairs 12, 13, 14, and 15.

Notes:

4. 复指数、余弦和正弦的变换
Transforms of Complex Exponentials, Cosines, and Sines

例：求证
Example. Prove that

$$e^{i\omega_0 t} \overset{\mathcal{F}}{\longleftrightarrow} 2\pi\delta(\omega-\omega_0).$$

由式 $(*)$，
Using identity $(*)$,

$$\begin{aligned}\mathcal{F}[e^{i\omega_0 t}] &= \int_{-\infty}^{+\infty} e^{i\omega_0 t}e^{-i\omega t}\,\mathrm{d}t \\ &= \int_{-\infty}^{+\infty} e^{-i(\omega-\omega_0)t}\,\mathrm{d}t \\ &= 2\pi\delta(\omega-\omega_0).\end{aligned}$$

类似地，
Similarly,

$$\begin{aligned}\mathcal{F}[\cos\omega_0 t] &= \mathcal{F}\left[\frac{e^{i\omega_0 t}+e^{-i\omega_0 t}}{2}\right] \\ &= \frac{1}{2}\mathcal{F}[e^{i\omega_0 t}] + \frac{1}{2}\mathcal{F}[e^{-i\omega_0 t}] \\ &= \frac{1}{2}\cdot 2\pi\delta(\omega-\omega_0) + \frac{1}{2}\cdot 2\pi\delta(\omega+\omega_0) \\ &= \pi\big[\delta(\omega+\omega_0)+\delta(\omega-\omega_0)\big].\end{aligned}$$

$$\mathcal{F}[\sin\omega_0 t] = i\pi\big[\delta(\omega+\omega_0)-\delta(\omega-\omega_0)\big].$$

Notes:

5. 单位阶跃函数的变换

Transform of the Unit Step Function

$$u(t)=\begin{cases}1, & t>0,\\ 0, & t<0.\end{cases}$$

可以验证

We can verify that

$$u(t)\ \overset{\mathcal{F}}{\longleftrightarrow}\ \frac{1}{i\omega}+\pi\delta(\omega).$$

从正变换形式上写出

The forward transform can formally be written as

$$\mathcal{F}[u(t)]=\int_{-\infty}^{+\infty}u(t)e^{-i\omega t}\,\mathrm{d}t=\int_{0}^{+\infty}e^{-i\omega t}\,\mathrm{d}t.$$

原稿接着用反变换验证：

The original notes then verify the result using the inverse transform.

$$\begin{aligned}\mathcal{F}^{-1}\left[\frac{1}{i\omega}+\pi\delta(\omega)\right]&=\frac{1}{2\pi}\int_{-\infty}^{+\infty}\left[\frac{1}{i\omega}+\pi\delta(\omega)\right]e^{i\omega t}\,\mathrm{d}\omega\\&=\frac{1}{2\pi}\left[\int_{-\infty}^{+\infty}\frac{e^{i\omega t}}{i\omega}\,\mathrm{d}\omega+\int_{-\infty}^{+\infty}\pi\delta(\omega)e^{i\omega t}\,\mathrm{d}\omega\right]\\&=\frac{1}{2\pi}\left[\int_{-\infty}^{+\infty}\frac{e^{i\omega t}}{i\omega}\,\mathrm{d}\omega+\pi\right].\end{aligned}$$

其中

where

$$\begin{aligned}\int_{-\infty}^{+\infty}\frac{e^{i\omega t}}{i\omega}\,\mathrm{d}\omega&=\int_{-\infty}^{+\infty}\frac{\cos\omega t+i\sin\omega t}{i\omega}\,\mathrm{d}\omega\\&=0+\int_{-\infty}^{+\infty}\frac{\sin\omega t}{\omega}\,\mathrm{d}\omega\\&=\begin{cases}\pi, & t>0,\\ -\pi, & t<0.\end{cases}\end{aligned}$$

注：由

Note. From

$$\int_{-\infty}^{+\infty} \frac{\sin x}{x}\,\mathrm{d}x = \pi$$

可得

we obtain

$$\int_{-\infty}^{+\infty} \frac{\sin kx}{x}\,\mathrm{d}x = \begin{cases} \pi, & k > 0, \\ -\pi, & k < 0. \end{cases}$$

代入上式，

Substituting into the preceding expression gives

$$\mathcal{F}^{-1}\left[\frac{1}{i\omega} + \pi\delta(\omega)\right] = \begin{cases} \dfrac{1}{2\pi}(\pi+\pi) = 1, & t > 0, \\ \dfrac{1}{2\pi}(-\pi+\pi) = 0, & t < 0, \end{cases} = u(t).$$

Notes:

6. 直流信号

A Constant (DC) Signal

直流信号 E 的变换对为

The transform pair for a constant (DC) signal E is

$$E \overset{\mathcal{F}}{\leftrightarrow} 2\pi E\delta(\omega).$$

$$\mathcal{F}[E] = \int_{-\infty}^{+\infty} Ee^{-i\omega t}\,\mathrm{d}t = 2\pi E\delta(\omega).$$

Notes:

二、傅里叶变换对的性质

Properties of Fourier Transform Pairs

教材 *P.190*。

textbook, p. 190.

约定

We use the notation

$$f(t) \overset{\mathcal{F}}{\leftrightarrow} F(\omega), \qquad g(t) \overset{\mathcal{F}}{\leftrightarrow} G(\omega).$$

全文采用

Throughout these notes, the transform convention is

$$F(\omega) = \int_{-\infty}^{+\infty} f(t) e^{-i\omega t}\,\mathrm{d}t, \qquad f(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} F(\omega) e^{i\omega t}\,\mathrm{d}\omega.$$

1. 性质 1：线性性

Property 1: Linearity

$$\begin{aligned}\mathcal{F}[af(t)+bg(t)] &= a\mathcal{F}[f]+b\mathcal{F}[g]\\ &= aF(\omega)+bG(\omega).\end{aligned}$$

或者

Alternatively,

$$\mathcal{F}^{-1}[aF(\omega)+bG(\omega)] = a\mathcal{F}^{-1}[F]+b\mathcal{F}^{-1}[G].$$

例 7.10

Example 7.10

$$\begin{aligned}\mathcal{F}[\sin^2 t] &= \mathcal{F}\left[\frac{1-\cos 2t}{2}\right]\\ &= \frac{1}{2}\left\{\mathcal{F}[1]-\mathcal{F}[\cos 2t]\right\}\\ &= \frac{1}{2}\left\{2\pi\delta(\omega)-\pi[\delta(\omega+2)+\delta(\omega-2)]\right\}.\end{aligned}$$

例 7.11

Example 7.11

设

Let

$$F(\omega) = \pi\delta(\omega+1) - \frac{i}{\omega+1},$$

求 $\mathcal{F}^{-1}[F(\omega)]$。

Find $\mathcal{F}^{-1}[F(\omega)]$.

解
Solution

$$\begin{aligned}\mathcal{F}^{-1}[F(\omega)] &= \frac{1}{2\pi}\int_{-\infty}^{+\infty}\left[\pi\delta(\omega+1)-\frac{i}{\omega+1}\right]e^{i\omega t}\,\mathrm{d}\omega \\ &= \frac{1}{2\pi}\left[\int_{-\infty}^{+\infty}\pi\delta(\omega+1)e^{i\omega t}\,\mathrm{d}\omega+\int_{-\infty}^{+\infty}\frac{-ie^{i\omega t}}{\omega+1}\,\mathrm{d}\omega\right] \\ &= \frac{1}{2\pi}\left[\pi e^{-it}+\int_{-\infty}^{+\infty}\frac{-ie^{i\omega t}}{\omega+1}\,\mathrm{d}\omega\right].\end{aligned}$$

令 $\omega+1=\xi$，则
Let $\omega+1=\xi$. Then

$$\begin{aligned}\int_{-\infty}^{+\infty}\frac{-ie^{i\omega t}}{\omega+1}\,\mathrm{d}\omega &= \int_{-\infty}^{+\infty}\frac{-ie^{i(\xi-1)t}}{\xi}\,\mathrm{d}\xi \\ &= e^{-it}\int_{-\infty}^{+\infty}\frac{-ie^{i\xi t}}{\xi}\,\mathrm{d}\xi \\ &= e^{-it}\begin{cases}\pi, & t>0,\\ -\pi, & t<0.\end{cases}\end{aligned}$$

代入得
Substitution gives

$$\begin{aligned}\mathcal{F}^{-1}[F(\omega)] &= \begin{cases}\dfrac{1}{2\pi}(\pi e^{-it}+\pi e^{-it}), & t>0,\\ \dfrac{1}{2\pi}(\pi e^{-it}-\pi e^{-it}), & t<0\end{cases} \\ &= \begin{cases}e^{-it}, & t>0,\\ 0, & t<0\end{cases} \\ &= u(t)e^{-it}.\end{aligned}$$

即
That is,

$$u(t)e^{-it} \overset{\mathcal{F}}{\leftrightarrow} \pi\delta(\omega+1)-\frac{i}{\omega+1}.$$

与
Compare this with

$$u(t) \overset{\mathcal{F}}{\leftrightarrow} \pi\delta(\omega)+\frac{1}{i\omega}$$

对比，频域发生了平移。
The comparison shows a shift in the frequency domain.

Notes:

2. 性质 2：平移性

Property 2: Shifting

$$f(t-t_0) \overset{\mathcal{F}}{\leftrightarrow} e^{-it_0\omega}F(\omega).$$

或者

Alternatively,

$$e^{i\omega_0 t}f(t) \overset{\mathcal{F}}{\leftrightarrow} F(\omega-\omega_0).$$

由正、逆变换公式可得以上关系。

These relations follow from the forward and inverse transform formulas.

例：由于

Example. Since

$$u(t) \overset{\mathcal{F}}{\leftrightarrow} \frac{1}{i\omega}+\pi\delta(\omega),$$

故

Hence,

$$u(t-c) \overset{\mathcal{F}}{\leftrightarrow} \left[\frac{1}{i\omega}+\pi\delta(\omega)\right]e^{-ic\omega}.$$

可进一步化简为

This can be simplified to

$$\frac{e^{-ic\omega}}{i\omega}+\pi\delta(\omega).$$

这里使用了 δ 函数的性质

Here we use the delta-function identity

$$\delta(x-x_0)\varphi(x)=\delta(x-x_0)\varphi(x_0).$$

另外，

In addition,

$$e^{-it}u(t) \overset{\mathcal{F}}{\leftrightarrow} \pi\delta(\omega+1)+\frac{1}{i(\omega+1)},$$

以及
and

$$\mathcal{F}^{-1}\left[\pi\delta(\omega-1)+\frac{1}{i(\omega-1)}\right]=u(t)e^{it}.$$

Notes:

3. 性质 3：相似性
Property 3: Scaling

若 $f(t)\overset{\mathcal{F}}{\leftrightarrow}F(\omega)$，则
If $f(t)\overset{\mathcal{F}}{\leftrightarrow}F(\omega)$, then

$$f(at)\ \overset{\mathcal{F}}{\leftrightarrow}\ \frac{1}{|a|}F\Big(\frac{\omega}{a}\Big),\qquad a\neq 0.$$

并且
and

$$\frac{1}{|a|}f\left(\frac{t}{a}\right)\ \overset{\mathcal{F}}{\leftrightarrow}\ F(a\omega).$$

例：求 $\mathcal{F}[u(5t-2)]$
Example: Find $\mathcal{F}[u(5t-2)]$

解（定义法）
Solution (using the definition)

$$\mathcal{F}[u(5t-2)]=\int_{-\infty}^{+\infty}u(5t-2)e^{-i\omega t}\,\mathrm{d}t.$$

令 $5t-2=\tau$，即 $t=(\tau+2)/5$，得
Let $5t-2=\tau$, so that $t=(\tau+2)/5$. We obtain

$$\begin{aligned}\mathcal{F}[u(5t-2)]&=\int_{-\infty}^{+\infty}u(\tau)e^{-i\omega(\tau+2)/5}\frac{1}{5}\,\mathrm{d}\tau\\&=\frac{1}{5}e^{-2i\omega/5}\int_{-\infty}^{+\infty}u(\tau)e^{-i(\omega/5)\tau}\,\mathrm{d}\tau\\&=\frac{1}{5}e^{-2i\omega/5}\left[\frac{1}{i(\omega/5)}+\pi\delta\Big(\frac{\omega}{5}\Big)\right]\\&=\frac{e^{-2i\omega/5}}{i\omega}+\frac{\pi}{5}\delta\Big(\frac{\omega}{5}\Big).\end{aligned}$$

注：利用 $\delta(\omega/5)=5\delta(\omega)$，还可将第二项写成 $\pi\delta(\omega)$。
note. Using $\delta(\omega/5)=5\delta(\omega)$, the second term can also be written as $\pi\delta(\omega)$.

Notes:

4. 性质 4：对称性

Property 4: Duality

若

If

$$f(t) \overset{\mathcal{F}}{\leftrightarrow} F(\omega),$$

则

then

$$F(t) \overset{\mathcal{F}}{\leftrightarrow} 2\pi f(-\omega).$$

证明

Proof

$$F(\omega) = \int_{-\infty}^{+\infty} f(t)e^{-i\omega t}\,\mathrm{d}t.$$

利用“定积分值与所用哑变量无关”，将上式中的 ω、t 互换：

A definite integral does not depend on the name of its dummy variable. Interchanging ω and t in the notation gives

$$F(t) = \int_{-\infty}^{+\infty} f(\omega)e^{-it\omega}\,\mathrm{d}\omega.$$

令 $\omega = -\xi$，

Let $\omega = -\xi$.

$$\begin{aligned} F(t) &= \int_{+\infty}^{-\infty} f(-\xi)e^{it\xi}\,\mathrm{d}(-\xi) \\ &= \int_{-\infty}^{+\infty} f(-\xi)e^{it\xi}\,\mathrm{d}\xi \\ &= \frac{1}{2\pi}\int_{-\infty}^{+\infty} [2\pi f(-\omega)]e^{it\omega}\,\mathrm{d}\omega. \end{aligned}$$

与反变换式比较，即得结论。

Comparison with the inverse transform formula proves the result.

例：由双边指数脉冲的变换对

Example. Start from the transform pair for a two-sided exponential pulse,

$$e^{-\beta|t|} \overset{\mathcal{F}}{\leftrightarrow} \frac{2\beta}{\omega^2+\beta^2}, \qquad \beta > 0,$$

可得
we obtain

$$\frac{2\beta}{t^2+\beta^2} \overset{\mathcal{F}}{\longleftrightarrow} 2\pi e^{-\beta|-\omega|} = 2\pi e^{-\beta|\omega|}.$$

因此
Therefore,

$$\frac{1}{t^2+\beta^2} \overset{\mathcal{F}}{\longleftrightarrow} \frac{\pi}{\beta} e^{-\beta|\omega|}.$$

原图说明：双边指数脉冲关于纵轴对称，在原点达到最大值，并向两侧衰减。
Description of the original figure. The two-sided exponential pulse is symmetric about the vertical axis, reaches its maximum at the origin, and decays in both directions.

注：$\beta > 0$ 是所列衰减指数变换对的适用条件，此处补明。
note. The condition $\beta > 0$ has been made explicit because it is required for the decaying-exponential transform pair.

Notes:

5. 性质 5：微分性
Property 5: Differentiation

(1) 时域微分
(1) Differentiation in the time domain

若 $f(\pm\infty) = 0$，则
If $f(\pm\infty) = 0$, then

$$f'(t) \overset{\mathcal{F}}{\longleftrightarrow} i\omega F(\omega).$$

证明：分部积分，
Proof. Integration by parts gives

$$\begin{aligned}\int_{-\infty}^{+\infty} f'(t)e^{-i\omega t}\,\mathrm{d}t &= f(t)e^{-i\omega t}\big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty}(-i\omega)f(t)e^{-i\omega t}\,\mathrm{d}t \\ &= 0 + i\omega\int_{-\infty}^{+\infty} f(t)e^{-i\omega t}\,\mathrm{d}t \\ &= i\omega F(\omega).\end{aligned}$$

(2) 频域微分
(2) Differentiation in the frequency domain

$$-itf(t) \overset{\mathcal{F}}{\longleftrightarrow} F'(\omega).$$

证明
Proof

$$F(\omega) = \int_{-\infty}^{+\infty} f(t)e^{-i\omega t}\,\mathrm{d}t,$$

$$\begin{aligned} F'(\omega) &= \int_{-\infty}^{+\infty} \frac{\partial}{\partial\omega}[f(t)e^{-i\omega t}]\,\mathrm{d}t \\ &= \int_{-\infty}^{+\infty} f(t)(-it)e^{-i\omega t}\,\mathrm{d}t. \end{aligned}$$

推广
Generalization

若
If

$$f(\pm\infty) = f'(\pm\infty) = \cdots = f^{(n-1)}(\pm\infty) = 0,$$

则
then

$$f^{(n)}(t) \overset{\mathcal{F}}{\leftrightarrow} (i\omega)^n F(\omega).$$

另一方面，
On the other hand,

$$(-it)^n f(t) \overset{\mathcal{F}}{\leftrightarrow} F^{(n)}(\omega).$$

由
From

$$\mathcal{F}[(-it)^n f(t)] = F^{(n)}(\omega)$$

可得
we obtain

$$\mathcal{F}[t^n f(t)] = i^n F^{(n)}(\omega).$$

特别地，
In particular,

$$\begin{aligned} \mathcal{F}[t^n \cdot 1] &= i^n[2\pi\delta(\omega)]^{(n)} \\ &= 2\pi i^n \delta^{(n)}(\omega). \end{aligned}$$

即
That is,

$$t^n \overset{\mathcal{F}}{\leftrightarrow} 2\pi i^n \delta^{(n)}(\omega).$$

Notes:

6. 广义傅里叶变换对举例

Examples of Generalized Fourier Transform Pairs

教材 *P.340*, 第 *16—32* 对中的若干项。

selected entries from pairs 16–32 on textbook p. 340.

右栏的导数均对 ω 求导。

All derivatives in the right-hand column are with respect to ω.

原编号 / Original no.	原函数 / Function	傅里叶变换 / Fourier transform
16	$u(t-c)$	$\left[\frac{1}{i\omega}+\pi\delta(\omega)\right]e^{-ic\omega}$
17	$t\,u(t)$	$i\frac{\mathrm{d}}{\mathrm{d}\omega}\left[\frac{1}{i\omega}+\pi\delta(\omega)\right]$
18	$t^n u(t)$	$i^n\frac{\mathrm{d}^n}{\mathrm{d}\omega^n}\left[\frac{1}{i\omega}+\pi\delta(\omega)\right]$
21	$u(t)e^{iat}$	$\frac{1}{i(\omega-a)}+\pi\delta(\omega-a)$
22	$u(t)e^{iat}t^n$	$i^n\frac{\mathrm{d}^n}{\mathrm{d}\omega^n}\left[\frac{1}{i(\omega-a)}+\pi\delta(\omega-a)\right]$
26	$\delta^{(n)}(t)$	$(i\omega)^n$
27	$\delta^{(n)}(t-c)$	$(i\omega)^n e^{-ic\omega}$
32	$t^n e^{iat}$	$2\pi i^n\delta^{(n)}(\omega-a)$

第 21、32 对由频域平移 $\omega\mapsto\omega-a$ 得到。

Pairs 21 and 32 follow from the frequency shift $\omega\mapsto\omega-a$.

Notes:

7. 性质 7：积分性

Property 7: Integration

性质编号沿用原稿，原稿此处未出现“性质 6”。

(1) 时域积分

(1) Integration in the time domain

若

If

$$\int_{-\infty}^{+\infty} f(t)\,\mathrm{d}t = 0,$$

则

then

$$\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau \overset{\mathcal{F}}{\leftrightarrow} \frac{F(\omega)}{i\omega}.$$

(2) 频域积分

(2) Integration in the frequency domain

$$\frac{f(t)}{-it} \overset{\mathcal{F}}{\leftrightarrow} \int_{-\infty}^{\omega} F(\xi)\,\mathrm{d}\xi.$$

证明（2）

Proof of (2)

令

Let

$$G(\omega) = \int_{-\infty}^{\omega} F(\xi)\,\mathrm{d}\xi,$$

求导得 $F(\omega) = G'(\omega)$。由微分性，

Differentiation gives $F(\omega) = G'(\omega)$. By the differentiation property,

$$\begin{aligned}\mathcal{F}^{-1}[F(\omega)] &= \mathcal{F}^{-1}[G'(\omega)] \\ &= -it\,\mathcal{F}^{-1}[G(\omega)].\end{aligned}$$

故

Hence,

$$\mathcal{F}^{-1}[G(\omega)] = \frac{\mathcal{F}^{-1}[F(\omega)]}{-it} = \frac{f(t)}{-it}.$$

即

That is,

$$\mathcal{F}^{-1}\left[\int_{-\infty}^{\omega} F(\xi)\,\mathrm{d}\xi\right] = \frac{\mathcal{F}^{-1}[F]}{-it}.$$

证明（1）

Proof of (1)

令

Let

$$g(t) = \int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau,$$

则 $f(t) = g'(t)$。在 $g(\pm\infty) = 0$ 时，由微分性，

Then $f(t) = g'(t)$. If $g(\pm\infty) = 0$, the differentiation property gives

$$\mathcal{F}[f(t)] = \mathcal{F}[g'(t)] = i\omega\mathcal{F}[g(t)].$$

从而

It follows that

$$\mathcal{F}[g(t)] = \frac{F(\omega)}{i\omega}.$$

注：若

Note. If

$$\int_{-\infty}^{+\infty} f(t)\,\mathrm{d}t \neq 0,$$

则

then

$$\mathcal{F}\left[\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau\right] = \frac{F(\omega)}{i\omega} + \pi\delta(\omega)F(0).$$

用 “卷积性” 易得。

This follows readily from the convolution property.

Notes:

三、作业

Assignments

11.22 作业

Assignment for November 22

- 教材 P.340，第 8 对的推导。
  Textbook, p. 340: derive transform pair 8.
- 教材 P.203，第 1 题（双）、第 2 题、第 3 题（双）。
  Textbook, p. 203: Exercise 1 (even-numbered parts), Exercise 2, and Exercise 3 (even-numbered parts).
- 教材 P.203，第 7 题的（4）、(5)、(6)。
  Textbook, p. 203: Exercise 7, parts (4), (5), and (6).

11.29 作业

Assignment for November 29

- 教材 P.203，第 4 题（双）。
  Textbook, p. 203: Exercise 4 (even-numbered parts).
- 教材 P.203，第 7 题的（1）、(2)、(3)。
  Textbook, p. 203: Exercise 7, parts (1), (2), and (3).
- 教材 P.203，第 6、8 题。
  Textbook, p. 203: Exercises 6 and 8.
- B 套，第 7 题。
  Set B, Exercise 7.

Notes: